

Next-Generation Agentic Fraud Prevention System

Strategic Requirements, Multi-Agent Architecture, and Technical Integration Specification

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1. Executive Summary

Traditional fraud detection infrastructure relies heavily on static, rule-based heuristics and isolated machine learning (ML) models operating in transactional silos. These legacy approaches fail to intercept complex fraud patterns—specifically Authorized Push Payment (APP) scams, social engineering coercion, SIM-swaps, and voice deepfake exploits—because they evaluate isolated actions against single thresholds without cross-channel context.

This document specifies the target architecture for an Agentic Fraud Prevention & Customer Loss Mitigation Platform. This system replaces static scoring with a network of autonomous, specialized AI agents operating on a shared, unified data plane. By orchestrating signal correlation, dynamic context enrichment, and interactive in-app customer verification within sub-2-second decisioning windows, the system intercepts fraud prior to authorization while minimizing false-positive friction for genuine customers.

2. Key Gaps in Current Systems vs. Agentic Architecture

Capability Gap	Legacy Banking Approach	Agentic Architecture Solution
Cross-Silo Signal Fusion	Detection tools run in departmental silos (Card, Wire, Online Banking, Call Center).	Unified Streaming Data Plane correlates telemetry across sessions, voice calls, device metrics, and rails in real time.

Capability Gap	Legacy Banking Approach	Agentic Architecture Solution
Interactive Intent Verification	Static SMS/Email OTPs or rigid push notifications easily bypassed via social engineering.	Conversational Intent Verification Agent interrupts scammer scripts via context-aware in-app dialogue.
Dynamic Threat Adaptation	ML models retrain on fixed weekly/monthly schedules; rules require manual policy changes.	Adversarial Red-Teaming Agent continuously probes controls with synthetic exploits and updates decision thresholds.
Pre-Authorization Orchestration	Systems block or approve blindly; lack nuanced, automated step-up workflows.	Multi-Agent Orchestration Engine executes bounded operational holds and dynamic step-up challenges within < 2 seconds.

3. Multi-Agent Network Architecture

The platform is structured into a three-tier model: a Unified Data & Context Layer, an Autonomous Multi-Agent Orchestrator, and a Bounded Execution & Governance Layer.

3.1 Core AI Agents Specifications

Agent 1: Signal Correlation & Perception Agent

Primary Objective: Detect behavioral anomalies and cross-channel threat indicators in real time.

Ingested Inputs: Mouse dynamics, typing cadence, IP/device velocity, SIM-swap status, active voice call state, location delta.

Autonomous Behavior: Synthesizes low-level telemetry to generate a continuous Coercion & Account Takeover (ATO) risk score. If a high-value wire transfer is initiated while a mobile device maintains an active phone call with an unrecognized number, this agent triggers an instant high-risk alert.

Agent 2: Dynamic Context & Graph Investigation Agent

Primary Objective: Enrich alerts with contextual background and graph relationships prior to taking action.

Ingested Inputs: Historical transaction graph, beneficiary account aging, consortium risk databases (known mule accounts), device relationship maps.

Autonomous Behavior: Automatically queries the customer relationship graph to determine whether the beneficiary account has structural ties to known fraud networks or displays patterns typical of newly created mule accounts.

Agent 3: Interactive Intent Verification Agent

Primary Objective: Counter social engineering by verifying customer intent directly within the banking channel.

Ingested Inputs: Coercion risk scores from Agent 1, customer communication preferences, active session state.

Autonomous Behavior: Bypasses passive SMS OTPs. Pauses transaction settlement and launches a dynamic in-app conversational dialogue using contextual prompts (e.g., 'Did a caller claiming to represent Tech Support instruct you to move these funds to a safe account?'), interrupting the scammer's psychological control.

Agent 4: Adversarial Red-Teaming Agent

Primary Objective: Continuously test internal controls against novel attack patterns.

Ingested Inputs: Emerging global fraud vectors, synthetic identity datasets, historical rule bypass logs.

Autonomous Behavior: Simulates novel attack scenarios against existing detection logic offline, automatically proposing threshold refinements and updating agent reasoning prompts prior to active exploit execution.

4. End-to-End Operational Workflow & Sequence

Step 1: Telemetry Capture & Anomaly Identification (Latency Budget: < 100ms)

The customer initiates a \$15,000 wire transfer via mobile app. The Signal Correlation Agent identifies that the device is on an active voice call with an unlisted number and detects hesitant typing behavior.

Step 2: Automated Graph Enrichment (Latency Budget: < 250ms)

The transaction details pass to the Dynamic Context Agent, which checks the target account against a distributed graph database. The beneficiary account was opened 48 hours prior and received three incoming payments from distinct originators within 4 hours.

Step 3: Contextual Risk Evaluation (Latency Budget: < 150ms)

The orchestrator aggregates signals from both agents. The system evaluates the situation as a high-probability Authorized Push Payment (APP) scam. Transaction settlement is placed on an immediate 15-minute operational hold.

Step 4: Dynamic Customer Intervention (Latency Budget: Active Channel)

The Interactive Intent Verification Agent takes over the app interface, presenting an interactive verification workflow with custom contextual prompts designed to challenge common scam scripts.

Step 5: Resolution & Audit Logging (Latency Budget: Post-Execution)

If the customer acknowledges coercion indicators, the transfer is canceled, the beneficiary account is reported to consortium databases, and the system issues a cryptographically signed Decision Token for compliance auditing.

5. Operational Governance & Risk Control Matrix

Risk Vector	Governance Constraint	Enforcement Mechanism
Agent Overreach	Autonomy boundaries strictly defined by financial exposure level.	High-impact actions (e.g., permanent account closure, regulatory SAR filing) require Human-in-the-Loop approval. Low-impact actions (temporary payment holds, step-up prompts) execute autonomously.
Regulatory Compliance & Explainability	Every automated decision must be fully reconstructible for regulatory review (e.g., CFPB, FCA).	All agent reasoning loops, tool calls, retrieved context, and evaluated weights are logged to an append-only ledger anchored by immutable Decision Tokens.
Model Drift & Hallucination	System decisioning must remain deterministic within policy parameters.	LLM reasoning components are bound to structured JSON outputs enforced by fixed Pydantic schemas and deterministic policy validation engines.

6. Technical Integration Specification & API Reference

The Interactive Intent Verification Agent acts as an intermediary between the Agentic Fraud Orchestration Engine and the mobile/web banking user interface. When the orchestrator identifies an elevated Coercion or APP scam risk, it halts transaction processing and issues an intervention request to this service.

6.1 Endpoint 1: Trigger Intent Verification Challenge

HTTP Request: POST /v1/interventions/evaluate | Content-Type: application/json

```
{
  "transaction_id": "tx_987654321_2026",
  "account_id": "acc_usr_883019",
  "session_id": "sess_44391209_abc",
  "risk_assessment": {
    "coercion_score": 0.89,
    "ato_score": 0.12,
    "primary_risk_factor": "ACTIVE_VOICE_CALL_WITH_SUSPICIOUS_MULE_BENEFICIARY"
  },
  "context_data": {
```

```

    "amount": 15000.00,
    "currency": "USD",
    "beneficiary_name": "Tech Support Services LLC",
    "beneficiary_account_age_days": 2,
    "device_telemetry": {
      "on_active_call": true,
      "call_duration_seconds": 840,
      "hesitant_typing_detected": true
    }
  }
}

```

HTTP Response (Challenge Required): 202 Accepted | Content-Type: application/json

```

{
  "intervention_id": "int_0091823746",
  "status": "CHALLENGE_REQUIRED",
  "hold_expires_at": "2026-08-07T12:53:09Z",
  "challenge_payload": {
    "ui_pattern": "CONVERSATIONAL_FRICTION_CARD",
    "primary_prompt": "We noticed you are currently on an active call while attempting this high-value wire transfer.",
    "warning_banner": "Scammers frequently pose as Tech Support, Bank Security, or Law Enforcement to trick you into moving funds.",
    "verification_questions": [
      {
        "question_id": "q1_coercion_check",
        "question_text": "Did someone over the phone instruct you to transfer this money to keep it 'safe'?",
        "type": "SINGLE_CHOICE",
        "options": [
          { "label": "Yes, I was told to move my money", "risk_weight": "HIGH" },
          { "label": "No, I personally know this recipient", "risk_weight": "LOW" }
        ]
      }
    ],
    "action_buttons": {
      "cancel_transaction": {
        "label": "Cancel Transfer & Lock Account",
        "action": "ABORT_AND_SECURE"
      },
      "proceed_verification": {
        "label": "Verify Identity via Live Video",
        "action": "BIOMETRIC_STEP_UP"
      }
    }
  }
}

```

6.2 Endpoint 2: Submit Customer Challenge Response

HTTP Request: POST /v1/interventions/int_0091823746/respond | Content-Type: application/json

```
{
  "intervention_id": "int_0091823746",
  "transaction_id": "tx_987654321_2026",
  "user_response": {
    "selected_question_id": "q1_coercion_check",
    "selected_option": "Yes, I was told to move my money",
    "user_action": "ABORT_AND_SECURE"
  },
  "client_metadata": {
    "timestamp": "2026-08-07T12:39:15Z",
    "call_active_during_response": true
  }
}
```

HTTP Response (Intervention Successful): 200 OK | Content-Type: application/json

```
{
  "decision": "TRANSACTION_CANCELLED",
  "action_taken": "ACCOUNT_PROTECTIVE_HOLD",
  "decision_token":
  "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ0eF9pZCI6InR4Xzk4NzY1NDMyMV8yMDI2Iiwicmlza19z
  Y29yZSI6MC44OSwiZGVjaXNpb24iOiJkQU5DRUxMRUQiLCJ0aW1lc3RhbnR4IiwiaWF0IjYMDI2LTA4LTA3VDEyOjM5Oj
  E1WiJ9...",
  "next_steps": {
    "ui_view": "SCAM_PREVENTION_SUMMARY",
    "message": "Your transfer was stopped to protect your account. A fraud specialist
  will contact you securely through the app."
  }
}
```

7. Cryptographic Decision Token Schema

```
{
  "header": {
    "alg": "RS256",
    "typ": "JWT",
    "kid": "fraud-agent-key-2026-v1"
  },
  "payload": {
    "iss": "agentic-fraud-core",
    "sub": "tx_987654321_2026",
    "account_id": "acc_usr_883019",
    "evaluating_agents": [
      "SignalCorrelationAgent_v4",
      "DynamicContextAgent_v2",

```

```
    "IntentVerificationAgent_v3"
  ],
  "final_decision": "DENIED_SCAM_COERCION",
  "confidence_level": 0.96,
  "iat": 1786106355,
  "exp": 1817642355
},
"signature": "RSA-SHA256 Signature (Private key bound to Hardware Security Module)"
}
```

8. Error Handling & Fallback Matrix

Error Code	Trigger Condition	System Action & Fallback
ERR_AGENT_TIMEOUT	Agent network fails to return decision within 500ms latency budget.	Transaction falls back to deterministic rule set; step-up verification defaults to standard 2FA challenge.
ERR_CLIENT_DISCONNECT	User closes application during active intent challenge.	Transaction remains on operational hold for the full 15-minute window; requires explicit manual sign-off to clear.
ERR_INVALID_TOKEN	Decision token fails cryptographic validation at authorization engine.	Transaction immediately blocked; security event logged for CISO alert platform.